

Assignment 3

due: 2026-09-30

For the action of G on itself by conjugation, the stabilizer $\text{Stab}_G(x)$ of $x \in G$ is usually written $C_G(x)$ and is called the *centralizer* of x in G . Note that

$$C_G(x) = \{y \in G \mid yxy^{-1} = x\} = \{y \in G \mid yx = xy.\}$$

Problem 1: Let $I = I_n$ be a finite set with n elements, and let $S = S_n = S(I_n)$ be the group of permutations of I . Recall that $|S| = n!$.

- Prove that there are $(n-1)!$ n -cycles in S . **Hint:** If the elements of I are written $I = \{a_1, a_2, \dots, a_n\}$, then the n -cycles $(a_1, a_2, \dots, a_n)$ and $(a_2, a_3, \dots, a_n, a_1)$ are *equal*.
- Prove that if σ is an n -cycle in S , then $C_S(\sigma) = \langle \sigma \rangle$

Last week you proved two results which you'll need to use in the next problem.

Proposition 0.1: Assume that H normalizes K . Then $\langle H, K \rangle = HK$. In particular, HK is a subgroup of G .

Proposition 0.2: If G is a finite group and H, K subgroups of G , then

$$|HK| = |H| \cdot |K| / |H \cap K|.$$

Problem 2: Let $\sigma, \tau \in S = S(\mathbb{Z}/n\mathbb{Z})$ be defined by: $\sigma(i) = i + 1$, $\tau(i) = -i$. Notice or recall $\tau\sigma\tau = \sigma^{-1}$, so that

$$D := \langle \sigma, \tau \rangle = \langle \sigma \rangle \langle \tau \rangle$$

is a subgroup of S (using the fact that $\langle \tau \rangle$ normalizes $\langle \sigma \rangle$, together with Proposition 0.1).

- Using Proposition 0.2, show that

$$|D| = |\langle \sigma \rangle| \cdot |\langle \tau \rangle| / |\langle \sigma \rangle \cap \langle \tau \rangle|.$$

What are $|\langle \sigma \rangle|$ and $|\langle \tau \rangle|$?

- Show that if n is odd, $\langle \sigma \rangle \cap \langle \tau \rangle = \{1\}$, and conclude $|D| = 2n$ in this case.
- Now suppose $n = 2k$ is even. Identify the unique subgroup of order 2 in $\langle \sigma \rangle$, and show that τ is **not** that element – i.e. $\sigma^k \neq \tau$. (Hint: evaluate both sides at a well-chosen point of $\mathbb{Z}/n\mathbb{Z}$.)
- Conclude that $\langle \sigma \rangle \cap \langle \tau \rangle = \{1\}$ for all n , and hence $|D| = 2n$ always. (D is usually written D_{2n} , the dihedral group of order $2n$.)

Problem 3: Let G be the group of invertible 2×2 matrices with entries in $F = \mathbb{Z}/p\mathbb{Z}$ for a prime number p ; the group operation is given by matrix multiplication.

- Show that $|G| = (p^2 - 1)(p^2 - p)$.
- Show that $T = \left\{ \begin{bmatrix} t & 0 \\ 0 & s \end{bmatrix} \mid t, s \in F^\times \right\}$ is a subgroup of G . Here $F^\times$ denotes the multiplicative group of invertible elements of $F = \mathbb{Z}/p\mathbb{Z}$. Also show that

$$T \text{ is isomorphic to } F^\times \times F^\times$$

- Show that $U = \left\{ \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \mid a \in F \right\}$ is a subgroup of G isomorphic to the additive group F .
- Show that T normalizes U . Find the order of the group $B = TU$.

Problem 4: Continue with G, T, U, B as in Problem 3.

A line in F^2 is by definition a linear subspace of dimension 1. For any non-zero vector v , the set $Fv = \text{Span}(v)$ is a line. Note that G acts in a natural way on the set of lines in F^2 .

- If we write $e = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $f = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ for the standard basis of F^2 , show that B is the stabilizer of the line Fv .
- Show that G acts transitively on the set of lines in F^2 .
- Conclude that the set of lines in F^2 is in bijection with the set G/B . How many lines are there in F^2 ?

Problem 5: For a group G and elements $x, y \in G$, the **commutator** of x and y is the element

$$[x, y] = x \cdot y \cdot x^{-1} \cdot y^{-1}.$$

- Let $G' = \langle [x, y], x, y \in G \rangle$ be the subgroup generated by all commutators. Prove that G' is a normal subgroup of G .
- Prove that the quotient group G/G' is **abelian (i.e. commutative)**.
- Suppose that H is any abelian group and that $f : G \rightarrow H$ is a group homomorphism. Prove that $\forall x \in G', f(x) = 1$; i.e. that $G' \subseteq \ker f$.
- Deduce that there is a homomorphism $\bar{f} : G/G' \rightarrow H$ for which $\bar{f} \circ \pi = f$ where $\pi : G \rightarrow G/G'$ is the quotient map.

Remark: G/G' is known as the **abelianization** of G .

Problem 6: Let Q be the subgroup of $\text{GL}_2(\mathbb{C})$ generated by the matrices

$$I = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, J = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}.$$

- Prove that Q is a group of order 8.
- Prove that the center Z of Q has order 2 and that

$$Q/Z \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

- Prove that $Z = Q'$ where Q' is the commutator subgroup.

Problem 7: Let F be any field and for $n \in \mathbb{N}, n > 0$, let $G = \mathrm{GL}_n(F)$. From linear algebra, we know that the determinant mapping

$$\det : G \rightarrow F^\times$$

is a group homomorphism, where $F^\times = F - \{0\}$ is the multiplicative group of the field.

Let $\mathrm{SL}_n(F) = \ker(\det)$ be the subgroup $\{g \in \mathrm{GL}_n(F) \mid \det g = 1\}$; it is known as the **special linear group**.

Prove that $\mathrm{GL}_n(F)/\mathrm{SL}_n(F) \simeq F^\times$.